http://ww1.microchip.com/downloads/en/devicedoc/doc1001.pdf

Switch S1 is used to turn on and turn off the power to the circuit. Capacitors C2 and C4 are used for suppressing high-frequency signals generated by the circuit. C7 and C8 are decoupling capacitors for crystal XTAL1.

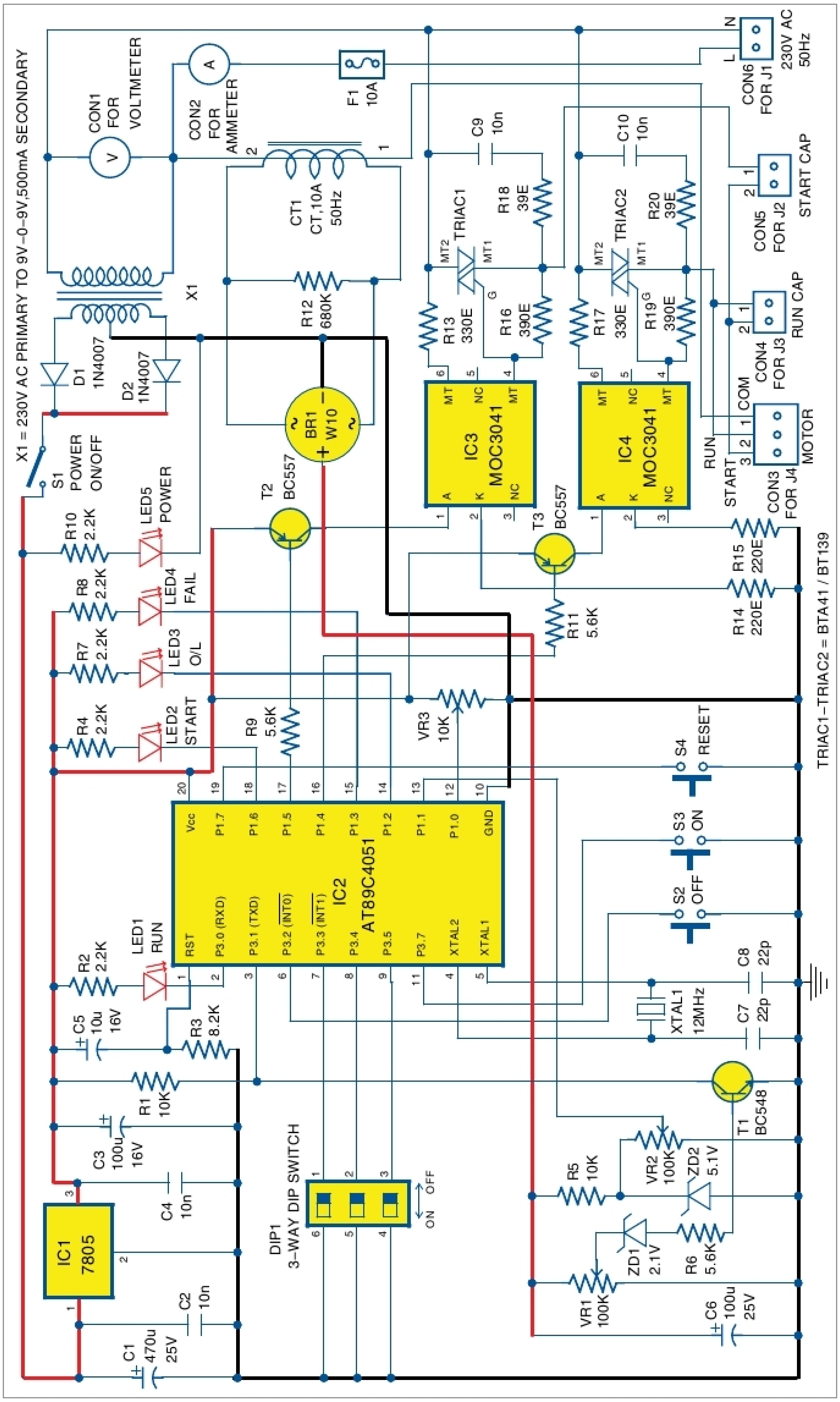

Fig. 2: Circuit diagram of submersible motor control panel

Capacitor C5 and resistor R3 form power-on reset circuit for MCU IC2. S2 is used to stop, and S3 to start the motor. S4 is used to reset the system from overload or motor failure condition.

PCB-mountable DIP switch (DIP1) is configured to get the required startup time delay. Total eight timing delays can be configured by adjusting switch positions. The table shows the delay against each configuration.

LEDs (LED1 through LED5) are used to indicate the various statuses of the system. Current transformer CT1 senses the motor current. Bridge rectifier BR1 rectifies CT1 output to DC, and C6 acts as a filter to smooth DC output of the rectifier.

VR1 can be used to set overload current value. A combination of ZD1, R6, T1 and R1 is used to provide motor current signal to the MCU.

One of the special characteristics of CT1 is that it burns out if connected without output load. Resistor R12 (680kΩ) connected across it is absolutely necessary to safeguard CT1.

Transistors T2 and T3 work as output drivers for optocouplers MOC3041. IC3 and IC4 optocouplers are used as triac drivers. IC3 and triac1 form the first solid-state relay (SSR1), and IC4 and triac2 form the second solid-state relay (SSR2).

Software

The software (subpump.c) is written in C language and compiled with Keil µvision 4 software. Logic behind the software is described briefly for better understanding.

One of the most important features of AT89C4051 is the inbuilt analogue comparator. This special feature of 8051 family chip is not utilised widely. In this project most of its special features, such as